INCH-POUND

MIL-PRF-1/690D
10 April 2007
SUPERSEDING
MIL-E-1/690C
1 May 1975

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, GAS

TYPE 5517

INACTIVE FOR NEW DESIGN
AFTER 21 JULY 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Rectifier, half-wave, cold cathode.

See figure 1.

Mounting position: Any.

Weight: 0.32 ounce (9.1 grams) nominal.

ABSOLUTE RATINGS:

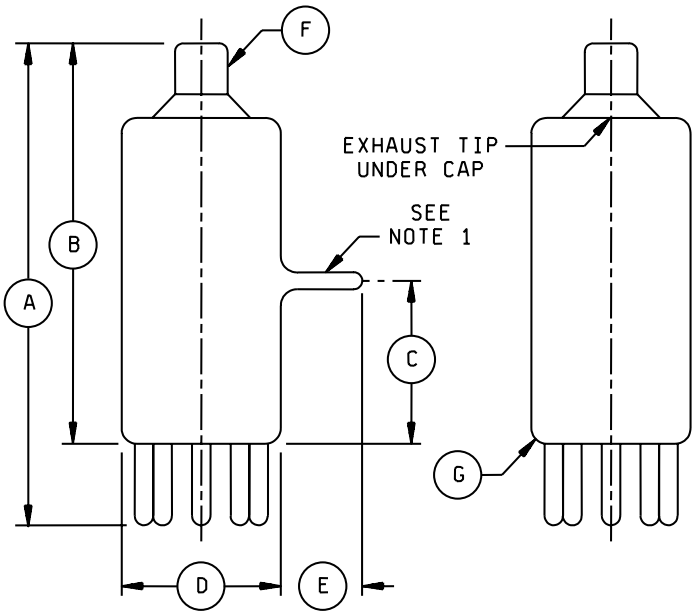
Parameter:	epx	Epp	C_ℓ	ib	R_ℓ	Io	i_{surge}	TA	Zp
Unit:	v	V ac	µF	mA	Ohms	mA dc	mA	°C	Ohms
Maximum:	2,800	---	---	100	---	12	300	+60	---
Minimum:	---	500	---	---	---	---	---	-50	5,000
Test conditions:	---	1,200 (see note 1)	2.0 ±10%	---	90,000 ± 1,000	---	---	---	---

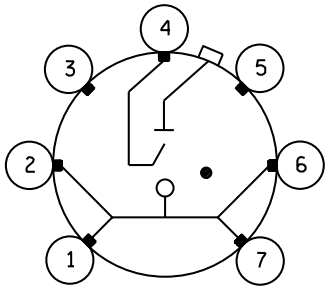
GENERAL:

Qualification: Not required.

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ALTERNATE OUTLINES





PIN CONNECTION
SCHEMATIC

Dimensions in inches				
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	---	2.250	---	57.15
B	1.780	1.970	45.21	50.04
C	.750	---	19.05	---
D	---	.750	---	19.05
E	---	.375	---	9.53
Conformance inspection, part 3 (see note 2)				
F	Cap: C1-3 (EIA)			
G	Base: E7-1 and envelope: 5-2 (T-5 1/2) (EIA)			

NOTES:
1. Exhaust tip will lie in sector between pins 1 and 7.
2. These dimensions shall be checked annually with an accept on zero defect sampling` plan using four test samples.

In case of failure after sampling, the falling dimension(s) shall become conformance inspection, part 2, for three successful consecutive submissions, at which time the test may revert to the conformance inspection, part 3 basis.

FIGURE 1. Outline drawing of electron tube type 5517.

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TABLE I. Testing and inspection.

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Acceptance level (see note 3)	Symbol	Limits		Unit
						Min	Max	
	Conformance inspection, part I							
1231	Peak emission by voltage drop	2/	ib = 100 mA	0.65	etd	---	125	v
1353	Operation of rectifiers	---	Zp = 6,000 ohms (min) (see figure 3)	0.65	Io	12.0	---	mA dc
	Conformance inspection, part 2							
1031	Low-frequency vibration	---	No voltages applied	---	---	---	---	---
3347	Ionization voltage	---	See figure 2	---	Epp	---	400	V ac
	Conformance inspection, part 3							
---	Life test	---	Group A; Zp/Io = 12.0 mA dc (min); t = 500 hours (see figure 3)	---	---	---	---	---
---	Life-test end points:	---		---				
1357	Operation of rectifiers	---		---	Io	11.0	---	mA dc
3347	Ionization voltage	---		---	Epp	---	450	V ac
1231	Peak emission by voltage drop	---		---	etd	---	150	v

1/ Epp shall be measured with the transformer unloaded, using a high-impedance voltmeter.

2/ With starter electrode connected to anode through a 10-megohm resistor.

3/ This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

Referenced documents. In addition to MIL-PRF-1, this document references MIL-STD-1311.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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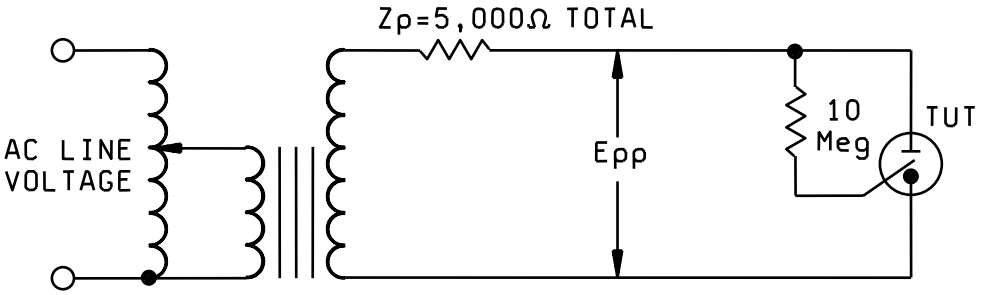


FIGURE 2. Ionization voltage test circuit.

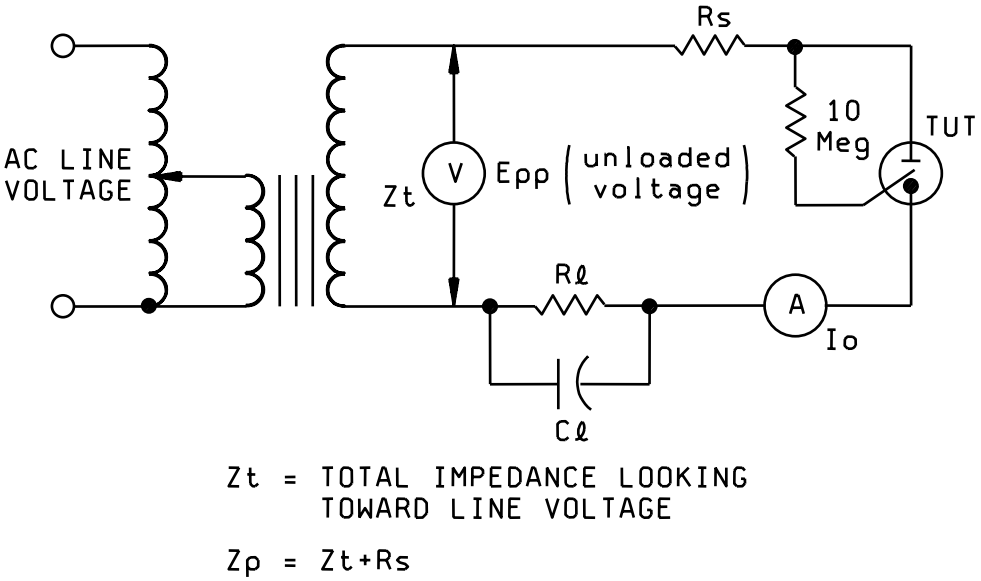


FIGURE 3. Life-test and operation of rectifiers test circuit.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2007-012)

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS, SH
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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